

Direct Hot-pressing Makes Sintering of Near-net-shape Parts Quick and Easy

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Abstract

A fast-heating sintering process developed by Dr. Fritsch has been employed in a wide range of industries for many years now. For instance, approx. 900 sintering presses based on this technology have been commissioned for the production of ceramics, diamond tools, friction pads as well as for other applications.

Introduction

Hot pressing is a well-known technique in the metal powder field. It is one of the sintering methods used to consolidate cold-pressed or loose powders.

Uniaxial hot pressing is mainly used in the ceramic and diamond tool business, although there are also many other applications. Sintering temperatures of 650 -2400 °C are normally used.

Within hot pressing technology, three distinctly different types of heating can be found in use:

- Inductive heating
- Indirect resistance heating
- Dr. Fritsch "DIRECT Hot-Pressing"

Inductive heating technology

In this process heat is produced within the mould when it is subjected to a high-frequency electromagnetic field, generated by using an induction coil coupled to an electronic generator. The mould is made out of graphite or steel, and pressure is applied by one or two cylinders onto the punches. The mould is positioned within the induction coil. The advantage here is that the pressure and the inductive power are completely independent. Even powders with a liquid phase are amenable to this process and low pressures are possible. The disadvantages include the expense of a high-frequency generator and the need for proper alignment. If the mould is placed off-centre, the heat distribution is uneven. But the main disadvantage is the dependence of the process on good inductive coupling and heat conductivity of the mould.

The magnetic field can penetrate the mould by only 0,5 mm to 3 mm. From there on, the heat has to be "transported" into the mould by the thermal conductivity of the mould material. Uniform heating is much more difficult if the air gap between the mould and the inductive coil is not the same all along the mould profile. Another potential problem is the heating rate. Too high a heat up rate will result in high temperature differences between the surface and core, which can destroy the mould.

Indirect resistance heating

With indirect resistance heating technology, the mould is placed in a heating chamber. The chamber is heated by graphite heating elements. These elements are heated by electric current. The heat is then transferred into the mould by convection. As the electrical energy heats the heating elements that then heat the mould in a secondary step, the process is called indirect resistance heating.

Advantages are high achievable temperatures, independent of the conductivity of the mould and independent of heat and pressure. Main disadvantage is the time that it takes to heat up the mould. It takes a relatively long time for heat transfer from the furnace atmosphere to the mould surface and subsequently throughout the cross-section of the mould.

Dr. Fritsch DIRECT Hot-Pressing

The basic idea of sintering with electric current going through the mould is quite old. Resistance heating of hard metal powders was patented by Tayler [1] as early as 1933. The compelling reason for shortening the cycle time then was to avoid grain growth and also to save energy. In DIRECT Hot-Pressing, the mould is directly connected to the electric power. The resistance of the mould and the powder part generates the heat directly in the mould. This results in very high heating rates. Additionally, this leads to a significant increase in the sintering activity of fine metal powder aggregates,

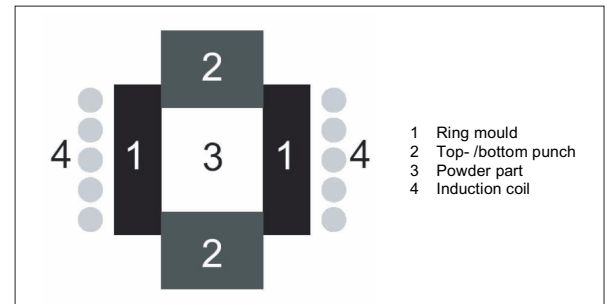


Fig. 1 Conventional inductive heating

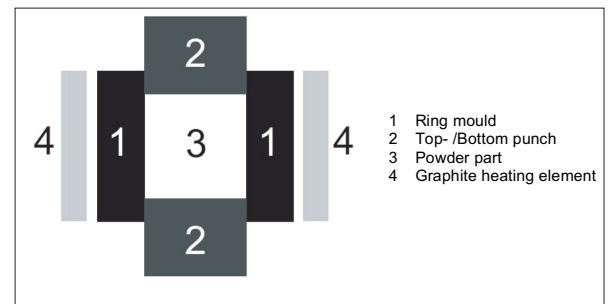


Fig. 2 Indirect resistance heating

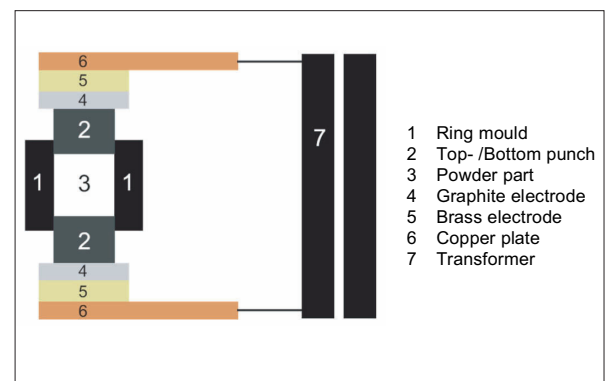


Fig. 3 Dr. Fritsch DIRECT Hot-Pressing

which makes short cycle times of a few minutes possible. Further, this process lowers the threshold sintering temperature and pressure compared to that required in conventional sintering processes. The previous two methods as illustrated in Fig. 1 and 2 are both closely dependent on an intrinsic property of the mould material, i.e. its thermal conductivity. With direct resistance heating, however, the heat is generated where it is needed. Recently, the manufacture of such critical



Fig. 4
DSP 518 sintering press

items as sputtering targets and high-performance ceramic components using B_4C , TiB_2 , $SiAlON$, etc., has been achieved. Using metal powder, the conductivity of the mould is ideal for fast heating of the workpiece. Moulds with a big diameter and relatively small height can be heated up very fast. The process is especially suitable for applications that need high heating rates, e.g. for materials that should not be kept at high tem-

peratures too long or for processes that require fast heating rates for high productivity.

With the DIRECT hot pressing technology, materials can be sintered to their final density. The near-net-shape precision is very high and in many cases saves mechanical reworking of the high-grade materials, which are often difficult to machine.

It can be used, for example, for the following materials and applications:

- Innovative hard metals
- Multi-layer and composite materials
- High-performance ceramic materials
- Alloys, FGMs
- Special materials to be sintered with minor grain growth
- Sputter targets
- Friction pads

Dr. Fritsch DIRECT Hot-Presses

Dr. Fritsch DIRECT Hot-Presses reach temperatures of up to 2400 °C and combine high heating rates with precise temperature control. The workpiece temperature is directly measured by means of a pyroscope or a thermocouple. One of the two

standard machines with type DSP 515 has a maximum pressing power of 600 kN (60 tonnes) applied on a sintering surface of max. 220 x 220 mm, heating power is 150 kVA at 4,5 V to 6 V secondary voltage. The machine can operate at vacuum or under protective atmosphere; over-pressure can also be realized. The machine has been successfully applied for metal and ceramic workpieces such as boron carbide at temperatures up to 2400 °C.

Process control and off-line programming by PC is also an option. The DSP 518 sintering press has a maximum sintering surface of 350 x 350 mm. Mould heights of up to 320 mm are possible. The standard configuration is equipped with an 180 kVA transformer and a pressing power of 1400 kN (140 tonnes). Every process parameter such as pressure, temperature, stroke and time are independently programmable. Fig. 5 shows a typical sintering program, displayed with the process monitoring software IPA-NT for Windows XP.

Literature

- [1] Tayler, G.F.: Apparatus for Making Hard Metal Compositions. U.S. Patent 1,896,854, 7 February 1933

Fig. 5
Process monitoring under Windows XP

Red/orange line:
actual/set temperature
Green line: densification of powder/green compact
Dark blue/light blue:
actual/set pressure

